

Deepfake-Eval-2024: A Multi-Modal In-the-Wild Benchmark of Deepfakes Circulated in 2024

Nuria Alina Chandra¹ Hannah Lee¹ Ryan Murtfeldt^{1,2} Lin Qiu^{1,2} Arnab Karmakar^{1,2}
Emmanuel Tanumihardja^{1,2} Kevin Farhat³ Ben Caffee^{1,2} Changyeon Lee⁴ Jongwook Choi⁵
Sejin Paik⁶ Aerin Kim^{1,7} Oren Etzioni^{1,2}

¹TrueMedia.org ²University of Washington ³Allen Institute for Artificial Intelligence
⁴University of Maryland ⁵Chung-Ang University ⁶Georgetown University ⁷Miraflow AI
alina.chandra11@gmail.com

Abstract

In the age of increasingly realistic generative AI, robust deepfake detection is essential for mitigating fraud and disinformation. While many deepfake detectors report high accuracy on academic datasets, we show that these academic benchmarks are out of date and not representative of real-world deepfakes. We introduce Deepfake-Eval-2024, a new deepfake detection benchmark consisting of in-the-wild deepfakes collected from social media and deepfake detection platform users in 2024. Deepfake-Eval-2024 consists of 45 hours of videos, 56.5 hours of audio, and 1,975 images, encompassing the latest manipulation technologies. The benchmark contains diverse media content from 88 different websites in 52 different languages. We find that the performance of open-source state-of-the-art deepfake detection models drops precipitously when evaluated on Deepfake-Eval-2024, with AUC decreasing by 50% for video, 48% for audio, and 45% for image models compared to previous benchmarks. We also evaluate commercial deepfake detection models and models finetuned on Deepfake-Eval-2024, and find that they have superior performance to off-the-shelf open-source models, but do not yet reach the accuracy of deepfake forensic analysts.

1. Introduction

Advances in generative AI models have precipitated a surge of highly realistic deepfakes, which have been used to fabricate messages from politicians [51], create non-consensual pornographic content [56], spread misinformation [20], and damage reputations [55], harming lives, businesses, and nations [5]. Between 2023 and 2024, there was a fourfold increase in the number of deepfakes detected in fraud [50], and in 2023 alone, an estimated 500,000 deepfakes were shared on social media websites [50].

Recent research has shown that people are no longer able to determine whether media is AI-generated or real [17]. Thus, the development of accurate and automated deepfake detection methods has become essential for mitigating the harmful effects of deepfakes. Many deepfake detection models have already been developed, such as GenConViT [62] for video, AASIST [24] for audio, and NPR [53] for image deepfakes, all of which perform extremely well on the academic datasets that they were originally benchmarked on, with AUC values approaching one (Table 3). However, these datasets are not representative of deepfakes circulating on social media, because many use outdated manipulation techniques (e.g., FaceForensics++ [47] and ForgeryNet [18]) with human differentiable fakes (e.g., faces not centered on heads in ForgeryNet). Further, most existing synthetic datasets and in-the-wild datasets have limited content diversity (e.g., exclusively single-scene videos containing a limited number of body poses [3, 12, 23], or exclusively English audio [25, 27, 33, 35]). In-the-wild deepfake are also limited, with the most recent prior datasets published in 2022 (audio) [35], and 2023 (video) [7], prior to widespread adoption of techniques such as stable diffusion [46] and the release of commercial models such as ElevenLabs voice conversion [14].

To address the limitations of existing deepfake detection benchmarks, we present Deepfake-Eval-2024, a dataset collected from social media and the free deepfake detection platform TrueMedia.org. Each item in Deepfake-Eval-2024 comes from a social media or TrueMedia.org user who flagged the media as potentially AI-manipulated in 2024. As a result, Deepfake-Eval-2024 is smaller than synthetic datasets, but much more diverse and directly representative of deepfakes in the wild. We summarize our contributions with the following:

- We release a challenging multimodal in-the-wild deepfake detection benchmark.
- To our knowledge, this is the first in-the-wild dataset that

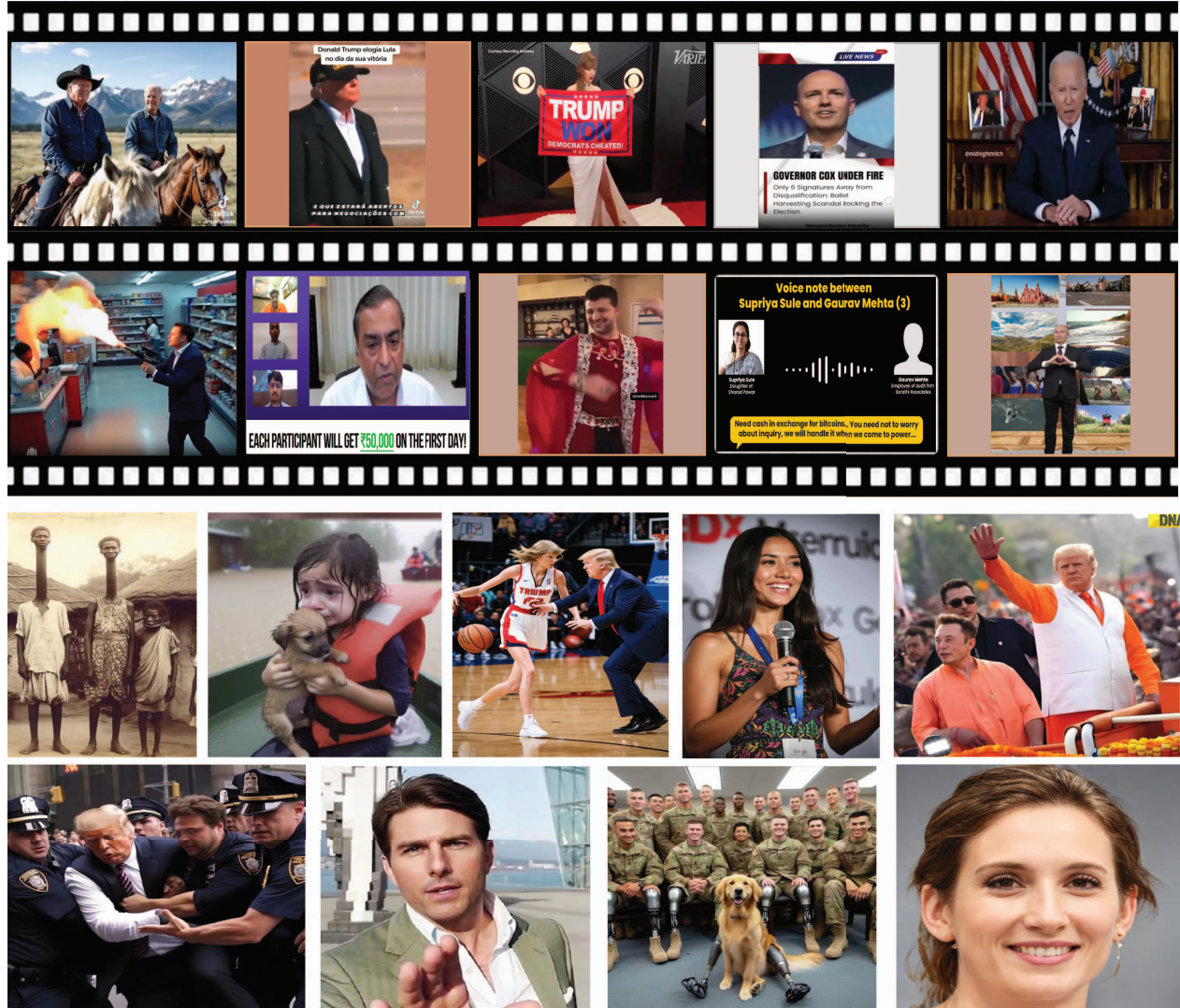


Figure 1. Examples of Deepfake-Eval-2024 video and audio (rows 1–2), and images (rows 3–4), demonstrating a diversity of content styles and generation techniques, including lipsync, faceswap, and diffusion. Images have been resized for presentation.

includes video, audio, and images, and it is the largest and most diverse in-the-wild deepfake detection dataset.

- We evaluate state-of-the-art deepfake detectors (both open-source and commercial) on contemporary in-the-wild data demonstrating their limitations and suggesting directions for future work.

2. Related Work

Deepfake datasets have not kept up with the fast-moving field of AI content generation. This is particularly true of in-the-wild datasets; prior to the release of Deepfake-Eval-2024, the only other in-the-wild audio deepfake dataset was published in 2022 [35], with fewer hours of audio than Deepfake-Eval-2024. To the best of our knowledge, prior to Deepfake-Eval-2024, there were no other in-the-

wild image-focused deepfake datasets. The most recent in-the-wild video datasets were from 2021 [41], and 2023 [7]. Supplementary Tables S1, S2, S3 provide a detailed survey of popular datasets that were released prior to Deepfake-Eval-2024.

Past in-the-wild datasets collect deepfakes by searching for media explicitly labeled as such (*e.g.*, [7] queried ‘deepfake’ in four languages), yet most real-world deepfakes posing security risks are unlabeled [19]. Deepfake-Eval-2024 instead draws from media flagged as potentially AI-manipulated by social media and TrueMedia.org users (primarily journalists), making it more representative of security-relevant deepfakes. It also vastly expands diversity: prior in-the-wild datasets used at most four sources [7], while Deepfake-Eval-2024 spans 88 web sources and

Table 1. Deepfake-Eval-2024 Summary Statistics

Modality	Size	Avg Resolution
Video	45.1 hrs	30.05 FPS, 576×720 px
Audio	56.5 hrs	44.66 kHz
Image	1,975 images	1,024×1,024 px

42 languages in our audio dataset and 52 different languages in combined video and audio datasets (Figure 2), compared to a maximum of two languages in prior audio datasets.

Most existing deepfake datasets are synthetically generated, enabling large scale (*e.g.*, ForgeryNet [18], DFDC [12], and AV-Deepfake1M [6]), but they fail to capture the distribution of deepfakes circulating on social media. Synthetic video datasets apply a limited set of manipulation techniques to curated, often highly structured real videos [3, 12, 23], focus exclusively on face manipulations, and image datasets largely repurpose video frames [47, 67] or target general AI-generated content rather than deepfakes [4, 61]. Deepfake-Eval-2024 addresses these gaps with diverse real-world content including varied settings, actions, and manipulations beyond the face (Figure 1).

3. Dataset

Deepfake-Eval-2024 is composed of 45 hours of videos and 56.5 hours of audio and 1,975 images. (Table 1 contains abbreviated summary statistics, with complete statistics stratified by label available in Supplementary Tables S4, S5, and S6.) The data includes real, AI-generated, and AI-manipulated content. Audio data includes audio from videos, in addition to audio-only media. The majority of video data has corresponding labeled audio.

An ideal benchmark for deepfake detection is representative of the real-world threat of deepfakes. This requires the following criteria: 1) it contains fake and real content that is difficult for humans to categorize, 2) it includes all popular generative techniques used for deepfake generation, and 3) it has diverse content, representative of media shared on the internet. We will show that Deepfake-Eval-2024 meets all three criteria through its unique data collection approach. All data was collected through the deepfake detection platform TrueMedia.org and social media content moderation forums.

3.1. Data Collection

Data Sources. The deepfake detection platform TrueMedia.org was a non-profit application used by journalists and fact-checkers starting in April 2024, and used by the general public starting in September 2024. Deepfake-Eval-2024 includes data uploaded to TrueMedia.org. Users provided a social media link or directly up-

loaded content to be checked for AI-manipulation. We also created a bot on X (previously known as Twitter) that allowed users to add content to our platform by tagging the bot. The code for this bot is available at <https://github.com/truemediaorg/socialbot>. In addition, we uploaded X posts that had been flagged by X Community Notes as potentially manipulated media. The top five most common data sources of Deepfake-Eval-2024 are X, direct upload, TikTok, Instagram, and Youtube (Figure 2, Supp. Figure S1).

Collection Ethics. TrueMedia.org users were informed in the Terms of Use that ‘by sharing your content, you agree to follow our content terms and you agree everything you share is consistent with these terms. Anything you share will not be private and can be used by us, our partners, and others we work with, in lots of different ways.’ To the best of our knowledge the dataset does not contain any personal identification information (*i.e.* social security numbers, emails, financial information, or medical records), and due to the nature of our data (non-textual), it is highly unlikely that we have incidentally collected such information through either TrueMedia.org or other web-sources.

Data Attributes. Our data collection method ensures that Deepfake-Eval-2024 is a **challenging** dataset. Users often brought media to TrueMedia.org when it could not be easily identified as real or fake by a human. Thus, we estimate that Deepfake-Eval-2024 has a greater proportion of challenging examples in both real and fake categories than prior datasets.

Dataset Diversity. Collecting data through social media and deepfake detection platform users also provides increased diversity with respect to generative models, ethnicity, language, and content. Deepfake-Eval-2024 is a sample of currently circulating AI-generated content. Thus, we estimate that our dataset includes AI-generated and manipulated content from every type of commonly used contemporary deepfake generation model. Further, TrueMedia.org users came from all over the world, resulting in increased ethnic and linguistic diversity in our dataset. Our dataset is 78.7% English and includes a total of 52 different languages (Figure 2). We estimate racial demographics by labeling 10% of the visual data for perceived race/ethnicity. Visual data is estimated to be 54.6% white and 45.4% non-white with full results in Supp. Table A.2.1. The content of the media itself has larger variations than the standardized content typically found in academic datasets. The diversity in data origins (Figure 2, Supp. Figure S1) results in a wide variety of different media styles, including videos of political speeches and self-shot content-creators, images of large crowds and close-up portraits, and audio clips of debating politicians.

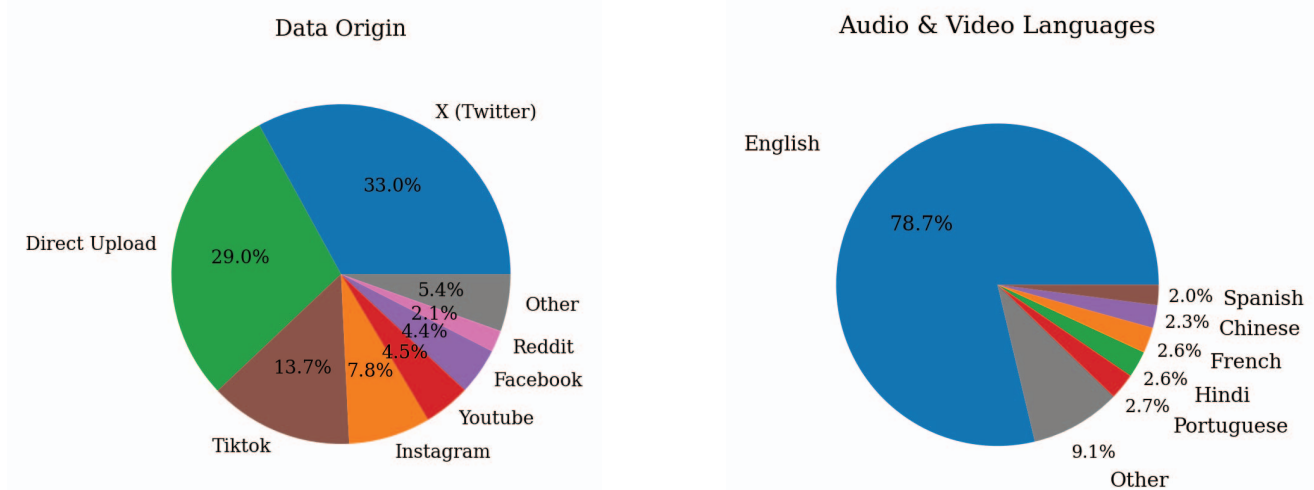


Figure 2. Distribution of data origins and languages in Deepfake-Eval-2024. In total, media was shared from 88 different web-domain names. The dataset also contains a total of 52 different languages (42 languages in Deepfake-Eval-2024-audio and 49 languages in Deepfake-Eval-2024-video). Languages were identified using speech recognition model Whisper [43].

3.2. Data Filtering

We remove duplicate data using a combination of manual review and hash functions. We include cases where two pieces of media have minor variations and thus appear to be the same (*e.g.*, different cropping of the same video). In order to tailor our datasets to evaluate deepfake detection models, we remove images and videos that do not contain photorealistic faces. This resulted in the removal of cartoons, art, and scenes without humans. We use GPT-4o (version 2024-08-06) to identify images with photorealistic faces. We note that GPT-4o’s responses have high precision but also a high false negative rate. To account for this, we manually review all images marked as non-photorealistic faces by GPT-4o. To identify videos with photorealistic faces we first use the dlib face detection library [28] to determine whether each frame contains a face or not. Videos where no faces are detected are then reviewed manually to check for missed faces.

3.3. Data Labeling

Label Definition. We label media as fake if it was AI-generated or manipulated. We choose to define our labels this way despite the challenge of differentiating AI-manipulated content from traditional forms of manipulation so that this dataset can be used to benchmark deepfake detection models which are trained to differentiate between AI-generated and real content.

Labeling Methodology. The labeling team consisted of seven people conducting deepfake forensic analysis: three experienced AI-generated content labelers and four machine learning research interns. The team met regularly to discuss the taxonomy, verification process, and edge

cases. Our verification process consisted of locating original sources using reverse image search or searching the web using quotes or situation descriptions, then confirming the trustworthiness of the source, and scanning media for characteristics of AI-generated media (Appendix B). See Appendix C for our detailed verification process. The team labeled each piece of media as “fake,” “real,” or “unknown” when the appropriate label was not clearly discernible from authenticated sources or media characteristics. Media labeled “unknown” were excluded.

Deepfake Forensic Analysis. We rely on articles published by professional fact-checking organizations such as Snopes [49] and AFP Fact Check [1] to confirm if AI manipulation was used. Additionally, we utilize community moderation platforms like X Community Notes to locate and review primary sources. We conduct source context verification by comparing social media posts with original materials such as full-length videos to verify if media has been manipulated. We further scrutinize the media for evidence of AI manipulation using specific forensic markers. For example, we rely on the synchronization of mouth movements with vocal sounds as a primary measure of authenticity in video and audio media. For videos and images, face-swaps are classified as fake if they were created after 2023, and unknown otherwise. This time separation was chosen based on research that suggests that the majority of face-swaps created in 2023 or later use AI [21]. Other common indicators of AI manipulation, including anatomical implausibilities, sociocultural implausibilities, and stylistic artifacts, are identified based on the framework created by Kamali et al. [26]. When the label of audio media cannot be determined from sources, due to the challenge of differ-

Table 2. Inter-rater disagreement statistics

Modality	N Checked	Total Disagreement	Real vs Fake	Real vs Unknown	Fake vs Unknown
Video	243	6.6%	2.1%	3.7%	0.8%
Audio	342	7.9%	0.6%	3.5%	3.8%
Images	269	9%	0%	3%	6%

entiating AI generated audio from non-AI generated voice impersonators, media is marked as fake if and only if there are both audible traits indicating that it is fake (*e.g.* socio-cultural implausibilities), in addition to at least two commercial audio detectors predicting the media as fake. Audio detectors alone are never used to label. We note that this labeling approach results in audio labels that are correlated with existing detectors, which is a limitation of the audio dataset. However, this approach is necessary, as it is impossible to differentiate between socioculturally-implausible audio made by highly skilled impersonators (real audio) and AI-generated audio (see Jordan Peele’s highly accurate Barack Obama impersonation which uses real audio and AI-manipulated video as an example [40, 45]). See Appendix B for complete labeling taxonomy and media examples.

Labeling Accuracy. To assess the consistency and quality of our annotations, the labeling team lead double-reviewed a random sample of 10% of the data. Annotations created by the team lead were excluded to avoid self-assessment bias. For videos, we find a 6.6% disagreement between labelers, with the largest discrepancy between real and unknown at 3.7%. For audio, we find a 7.9% disagreement between labelers, with the largest discrepancies between real and unknown at 3.5% and between fake and unknown at 3.8%. And for images, we find a 9% disagreement between labelers, with the largest discrepancy between fake and unknown at 6%. (See Table 2 for complete disagreement breakdown.) This disagreement between trained labelers represents the challenge of the task. Given that the inter-labeler disagreement was consistently below 10%, we posit that deepfake detection models should be able to achieve at least 90% accuracy on Deepfake-Eval-2024, and likely higher given that real vs. fake disagreement is always below 2.5%.

Common labeling errors include: differentiating between dubbed videos (where the video has not been AI-manipulated, and thus is real), and lipsynced videos (where the video has been AI-manipulated to make the mouth match new words and thus should be marked as fake); determining if audio sound is synthetic; and missing anatomical or sociocultural implausibilities.

4. Experiments

Model Selection. To evaluate the state-of-the-art of deepfake detection on real world in-the-wild deepfakes, we test an array of open-source deepfake detection models on Deepfake-Eval-2024. We select standard models that encompass the primary deepfake detection model architectures associated with each modality. Models were also selected based on the availability of pretrained model weights and runnable training code. All models were chosen prior to experimentation, and no models were omitted on the basis of performance.

Open-Source Models. For each modality, we evaluate three different open-source deepfake detection models on the modality-appropriate Deepfake-Eval-2024 data. For image detection we include a single layer perceptron with a CLIP [42] backbone (UFD [39]), a model based on diffusion inversion (DistilDIRE [32]), and a convolutional neural network (NPR [53]). For audio detection we include a spectro-temporal graph attention network (AASIST [24]), a convolutional neural network applied to raw waveforms (RawNet2 [52]), and a model with a self-supervised component (P3 from Wang et al. [58]). We choose video models that have a generative convolutional vision transformer (GenConViT [62]), a temporal convolutional network (FTCN [66]), and a model that evaluates style latent vectors (Styleflow [8]). We also evaluated all open source multimodal models with runnable code that we could find, AVF [15] and FGI [2]. We use the code and preprocessing approaches described in the original publications. To adapt to open-source audio models with a limit of four seconds, we split Deepfake-Eval-2024 audio files into four-second segments and report performance on these segments.

Commercial Models. We evaluate commercially available deepfake detection models from companies that partnered with TrueMedia.org: Hive, Reality Defender, Pindrop, AI or Not, Hiya, Fraunhofer, and Sensity AI. Many companies provide multiple models. In total, we evaluate 22 different commercial models (six video, eight audio, and eight image models). We anonymize the performance of the models and providers to comply with contractual agreements. Commercial models are evaluated using their latest available versions as of December 2024. All vendors were blind to the test data. Due to the high per-query cost of commercial vendors, we were unable to evaluate all commercial

models on the entirety of Deepfake-Eval-2024. Instead, we evaluate all models on a subset of Deepfake-Eval-2024, and then evaluate the top three models for each modality on the entire Deepfake-Eval-2024. We report the performance of the top commercial model for each modality in Table 5, and Supp. Table S10.

Evaluation Metrics. Past deepfake detection publications have selectively focused on specific metrics (*e.g.* AUC in [64], accuracy in [39], and EER in [60]). Selective reporting on specific metrics makes comparison across publications challenging and does not offer a comprehensive view of model performance. As such, we evaluate the performance of models on Deepfake-Eval-2024 through a wide variety of metrics focusing on AUC, F1-score, and accuracy, with full metrics including precision, recall, false positive rate, and false negative rate available in the supplementary materials. We also report EER for audio models in the supplementary, as it is common in audio deepfake literature. Some open-source and commercial models fail to run on all media files due to model constraints (*e.g.*, media length limits, or requirements for a face to be detected in a certain number of frames). When a model fails to produce a prediction, we exclude this file when calculating the metrics for the associated model.

Evaluation on Previous Benchmarks. We compare the performance of open-source models on our dataset to the performance of each model on the test datasets reported in its original publication (Table 3). To account for different reporting metrics used across publications, we recompute predictions on the originally published test datasets to provide a full array of evaluation metrics. Where multiple test datasets were reported in the original publication, we compute results on as many of the datasets as possible and report average metrics across these test datasets.

Finetuning. To determine if real-world deepfake detection performance can be improved by training on representative data, we finetune all open-source models on 60% of Deepfake-Eval-2024, and evaluate the performance on the remaining 40% of the data (Table 4). This split mirrors real-world scenarios where models must generalize from limited training data to detect unseen deepfake techniques. We finetune each model following the original authors' recommended training procedures and hyperparameters where available, using early stopping to avoid overfitting.

4.1. Open-Source Model Performance

All off-the-shelf open-source models perform poorly on Deepfake-Eval-2024. The maximum AUC of open-source models across modalities and models was 0.58 (Table 3, Supp. Table S8). The performance of multimodal models is similarly poor (Supp. Table S11). Further, many off-the-shelf models have an AUC close to 0.5, the same as random guessing, suggesting that these models perhaps learned to

predict deepfakes based on correlations that were present in academic training datasets but do not exist in contemporary real-world data.

Performance on Deepfake-Eval-2024 is considerably lower than on previous benchmarks. The poor performance of open-source models on Deepfake-Eval-2024 offers a stark contrast to the exceptional performance of these models on the datasets that they were originally tested on (right side of Table 3). We observe an average drop in AUC of 50% for video, 48% for audio, and 45% for image models when evaluated on Deepfake-Eval-2024, as compared to the academic datasets that the models were originally tested on. This drastic difference in performance suggests that the academic deepfake detection datasets which the models were trained to perform well are not representative of the threat of contemporary deepfakes, underscoring the importance of up-to-date, challenging, in-the-wild deepfake datasets like Deepfake-Eval-2024.

4.2. Finetuned Model Performance

Models improve when finetuned on a subset of Deepfake-Eval-2024. AUC for open-source models improves by an average of 57.6% for video, 80.6% for audio and 4.5% for images (Table 4, Supp. Table S9). However, the degree of improvement varies across models, suggesting that some model architectures may be less suited to adapt to the challenges of real-world deepfake detection. For example, the simple single-layer UFD model learns to predict all data as fake after finetuning, and video model Styleflow also shows limited improvement in AUC. The limited improvement in image models is likely attributable to the relatively small amount of image finetuning data, which was insufficient to shift the highly parameterized models of NPR and DistilDIRE towards the distribution of in-the-wild deepfakes. Although performance improves after finetuning, there is still significant room for further improvement, with the peak accuracy reaching 0.75 for videos, 0.86 for audio, and 0.63 for images, which is below the 90% lower bound estimate of human deepfake analyst accuracy (Section 3.3). These results suggest that in addition to more representative training datasets, new model paradigms may be needed for robust and reliable deepfake detection.

4.3. Commercial Model Performance

Top commercial models exceed the performance of open-source models. Top commercial models considerably outperform off-the-shelf open-source models and finetuned image models, and perform slightly better than finetuned audio and video models. Commercial audio model performance is particularly strong, which is likely attributable to the limitations of the audio labeling approach as described in Section 3.3. No commercial models that we evaluated had an accuracy of 90% or above, suggesting that com-

Table 3. Open-source model performance on Deepfake-Eval-2024 and original benchmarks

Modality	Model	Deepfake-Eval-2024				Original Publication Test Data			
		AUC	Prec.	Recall	F1	AUC	Prec.	Recall	F1
Video	GenConViT [62]	0.63	0.60	0.50	0.54	0.96	0.93	0.99	0.96
	FTCN [66]	0.50	0.51	0.67	0.41	0.87	0.91	1.00	0.95
	Styleflow [8]	0.51	0.54	0.43	0.48	0.95	0.96	0.89	0.77
Audio	AASIST [24]	0.43	0.31	0.51	0.39	1.00	1.00	0.95	0.97
	RawNet2 [52]	0.53	0.66	0.39	0.49	0.99	0.60	0.99	0.74
	P3 [58]	0.58	0.36	1.00	0.53	1.00	1.00	0.96	0.98
Image	UFD [39]	0.56	0.63	0.999	0.77	0.94	0.95	0.67	0.75
	DistilDIRE [32]	0.52	0.64	0.87	0.74	0.99	0.99	0.98	0.98
	NPR [53]	0.53	0.69	0.29	0.41	0.98	0.95	0.94	0.94

Original publication test data includes the following datasets for each model. Where multiple datasets are specified, the reported metrics are averages over these datasets. Evaluations were re-run on original publication test data if metrics were not provided in the original publication. GenConViT: [47], [12], [31], [29]; FTCN: [31]; Styleflow: [31], [13], [30], [22]; AASIST, RawNet2, and P3 were all evaluated on the LA eval set of ASVspoof2019 [59]; UFD: [57] and subsets of LAION-400M [48] and AI generated images from latent diffusion models [46], Glide [38], and DALL-E mini [10] provided by the original publication [39]; DistilDIRE: ImageNet and AI generated images from Stable Diffusion v1 [46] and ADM [11] as specified in the original publication [32]; NPR: [54], [61], and the dataset from [39]. Complete performance metrics on Deepfake-Eval-2024 are available in Supp. Table S8.

Table 4. Open-Source Model Finetuning Results. Complete performance metrics are available in Supp. Table S9.

Modality	Model	Accuracy	AUC	F1
Video	GenConViT [62]	0.75	0.82	0.71
	FTCN [66]	0.65	0.71	0.62
	Styleflow [8]	0.53	0.56	0.58
Audio	AASIST [24]	0.84	0.91	0.78
	RawNet2 [52]	0.82	0.88	0.86
	P3 [58]	0.86	0.92	0.81
Image	UFD [39]	0.63	0.56	0.77
	DistilDIRE [32]	0.61	0.56	0.73
	NPR [53]	0.69	0.73	0.76

Table 5. Best Commercial Model Performance on Deepfake-Eval-2024

Modality	Accuracy	AUC	Precision	Recall	F1
Video	0.78	0.79	0.77	0.77	0.77
Audio	0.89	0.93	0.89	0.84	0.87
Image	0.82	0.90	0.99	0.71	0.83

mercial models still need improvement to reach the accuracy of human deepfake forensic analysts. In addition, we note that open-source models finetuned on a subset of Deepfake-Eval-2024 approach the accuracy of commercial models (finetuned video model GenConViT has an accuracy

of 75%, and finetuned audio model P3 from [58] has an accuracy of 86%). This suggests that the competitive advantage of these commercial models may be derived primarily from training dataset curation.

5. Error Analysis

Error analysis methodology. To further investigate detection model failures, we identify media traits associated with errors. We perform manual error analysis on the entire finetuning test set of videos and images. Due to the length of the Deepfake-Eval-2024 audio test set, we were unable to evaluate its entirety and instead we manually evaluate a class-balanced random sample of 10%, consisting of 2000 four-second audio clips. For each modality we identify media traits that are associated with a statistically significant decrease in accuracy as measured by chi-squared tests for each model separately, using significance threshold $p < 0.05$. We exclude models that predict all data as belonging to a single class from this error analysis (off-the-shelf and finetuned UFD [39] and off-the-shelf P3 [58]).

Off-the-shelf models perform worse on diffusion-generated videos. Off-the-shelf GenConViT and FTCN have an average 21.3% lower accuracy on videos which appear to be generated by a diffusion model (*e.g.*, OpenAI’s Sora). (Videos were identified as likely diffusion-generated through the presence of watermarks and diffusion-associated visual characteristics [26].) After finetuning on Deepfake-Eval-2024, which includes other diffusion videos, the accuracy gap narrows to 5.4%, suggesting

that the off-the-shelf models’ underperformance was primarily due to a domain shift.

Models are challenged by videos with selective facial manipulation and videos with non-facial manipulations. Deepfake-Eval-2024 includes atypical manipulation patterns such as videos with a mix of real and fake faces, and manipulations in non-facial regions. This differs from conventional datasets that contain either entirely real or entirely fake content. Videos with selective face manipulation where some faces are AI-manipulated while others remain real show a 31% decrease in accuracy. Videos with non-facial manipulation (*e.g.*, altered objects or locations, body modifications) experience a 17.4% decrease in accuracy compared to other videos. These performance drops likely stem from the fundamental design of most video detection models, which typically assume that the AI manipulation exists for the entire video, and that the fake areas are confined to faces. Consequently, even after finetuning, these types of manipulations remain particularly challenging for models to detect, with accuracy deficits of 16.85% for selective face manipulation and 35.47% for non-facial manipulation as compared to the complementary groups.

Models perform worse on audio with non-English languages, silences, and background noise, specifically music. We focus exclusively on the errors of finetuned audio models because floor effects associated with low performance (Supp. Table S8) make off-the-shelf performance indistinguishably poor across traits. In finetuned models, non-English audio has an average accuracy that was 7.21% lower than English audio. Because Deepfake-Eval-2024 is an in-the-wild dataset, some parts of the included audio files are silent. Models have 35.39% worse accuracy on audio clips that are silent. This is expected behavior, as audio without speakers is out of distribution for models targeting deepfake audio. We also note that models perform worse on audio with background noises (accuracy decrease of 7.66%). Music in the background is associated with a drop in accuracy of 17.94%, and a large increase of 26.12% in false negative rate. Adding background music is a common technique in deepfake generation, and our results suggest that current models often fail to identify fakes when the fakes have music added. The inability of models to accurately predict audio with music is a major vulnerability in existing audio deepfake detection models.

Models have lower accuracy on images with text overlays. We identify image categories such as depicted crowds, skin color, and the presence of text overlays. We did not find any statistically significant differences in performance associated with these categories, but we do observe a decrease in accuracy when images have text: accuracy of finetuned models decreases by 9% and the F1-score decreases by 10.5% on average. This indicates a distributional mismatch with existing training datasets, which do not include

images with text overlays. We hypothesize that the lack of statistically significant differences in the image models is due to floor effects, as the average accuracy of off-the-shelf and finetuned audio models was low (Supp. Tables S8, S9).

Many errors are not attributable to human-identifiable characteristics. The identified error-associated media characteristics do not encompass all errors in the dataset (*e.g.*, an average of 33% of audio errors have music in the background, and an average 5% of the image errors had text overlays). The errors not associated with media traits are caused by other failures in model signal interpretation. Developing methods to identify non-visible error patterns is an important area of future research.

6. Discussion & Limitations

In-the-wild deepfake data is crucial for evaluating detection models against real-world threats. However, we acknowledge that curating and manually labeling in-the-wild data is costly and susceptible to human error, resulting in a dataset size appropriate for evaluation but insufficient for wide-scale training. As such, there is still a need for synthetic deepfake datasets, and we recommend that these should strive to be more representative of real-world data and contain the media characteristics that our study reveals to be associated with model errors.

While Deepfake-Eval-2024 is the most comprehensive and diverse collection of real-world deepfakes available today, the rapid evolution of generative AI means that datasets can quickly become outdated, and more work is needed to develop systems that track emerging deepfakes and regularly update datasets. There is also a risk of adversarial actors using Deepfake-Eval-2024 to develop new deepfake generation techniques that evade detectors. To address this risk we make the dataset available at TrueMedia.org with a CC-BY-SA-4.0 license, but gate access to individuals who are verifiably working on deepfake detection or at research institutions (See Appendix A.5).

Ultimately, we believe the release of Deepfake-Eval-2024 as a benchmark for contemporary real-world deepfake detection will catalyze the development of robust models that can effectively address the evolving threat of modern deepfakes.

7. Acknowledgements

This work was made possible through the incredible team at TrueMedia.org, including Alex Schokking, Art Min, Dawn Wright, Field Cady, James Allard, Kathy Thrailkill, Maryvel Dolotanora, Maxwell Bennett, Michael Bayne, Michael Langan, Molly Norris Walker, Paul Carduner, and Steve Geluso. We would like to thank Ranjay Krishna and Ludwig Schmidt for their advice and Camp.org for the generous funding that supported this work.

References

- [1] Agence France-Presse. About afp fact check, 2025. AFP Fact Check is the fact-checking division of Agence France-Presse, a major world news agency. 4
- [2] Marcella Astrid, Enjie Ghorbel, and Djamila Aouada. Detecting audio-visual deepfakes with fine-grained inconsistencies. In *British Machine Vision Conference (BMVC)*, 2024. 5, 6
- [3] Sarah Barrington, Matyas Bohacek, and Hany Farid. Deepspk dataset v1.0, 2024. 1, 3, 2
- [4] Jordan J. Bird and Ahmad Lotfi. Cifake: Image classification and explainable identification of ai-generated synthetic images. *IEEE Access*, 12:15642–15650, 2023. 3
- [5] Tina Brooks, G. Princess, Jesse Heatley, J. Jeremy, Scott Kim, M. Samantha, Sara Parks, Maureen Reardon, Harley Rohrbacher, Burak Sahin, S. Shani, S. James, T. Oliver, and V. Richard. Increasing threats of deepfake identities. Public-private analytic exchange program report, U.S. Department of Homeland Security, 2021. 1
- [6] Zhixi Cai, Shreya Ghosh, Aman Pankaj Adatia, Munawar Hayat, Abhinav Dhall, Tom Gedeon, and Kalin Stefanov. Av-deepfake1m: A large-scale llm-driven audio-visual deepfake dataset. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 7414–7423, 2024. 3, 2
- [7] Beomsang Cho, Binh M. Le, Jiwon Kim, Simon Woo, Shahroz Tariq, Alsharif Abuadbba, and Kristen Moore. Towards understanding of deepfake videos in the wild. In *Proceedings of the 32nd ACM International Conference on Information & Knowledge Management (CIKM)*, UK, 2023. 1, 2
- [8] Jongwook Choi, Taehoon Kim, Yonghyun Jeong, Seungryul Baek, and Jongwon Choi. Exploiting style latent flows for generalizing deepfake video detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1133–1143, 2024. 5, 7
- [9] Hao Dang, Feng Liu, Joel Stehouwer, Xiaoming Liu, and Anil K Jain. On the detection of digital face manipulation. In *In Proceeding of IEEE Computer Vision and Pattern Recognition (CVPR 2020)*, Seattle, WA, 2020. 3
- [10] Boris Dayma, Suraj Patil, Pedro Cuenca, Khalid Saifullah, Tanishq Abraham, Phuc Le Khac, Luke Melas, and Ritobrata Ghosh. Dall-e mini, 2021. 7
- [11] Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in Neural Information Processing Systems*, 34:8780–8794, 2021. 7
- [12] Brian Dolhansky, Joanna Bitton, Ben Pfau, Jikuo Lu, Russ Howes, Menglin Wang, and Cristian Canton Ferrer. The deepfake detection challenge dataset, 2020. 1, 3, 7, 2
- [13] Nick Dufour and Andrew Gully. Contributing data to deepfake detection research. Google AI Blog, 2019. [Accessed 30-07-2023]. 7
- [14] ElevenLabs. About elevenlabs, 2025. Accessed on March 2, 2025. 1
- [15] Chao Feng, Ziyang Chen, and Andrew Owens. Self-supervised video forensics by audio-visual anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10491–10503, 2023. 5, 6
- [16] Joel Frank and Lea Schönherr. WaveFake: A Data Set to Facilitate Audio Deepfake Detection. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021. 3
- [17] Joel Frank, Franziska Herbert, Jonas Ricker, Lea Schönherr, Thorsten Eisenhofer, Asja Fischer, Markus Dürmuth, and Thorsten Holz. A representative study on human detection of artificially generated media across countries, 2023. 1
- [18] Yinan He, Bei Gan, Siyu Chen, Yichun Zhou, Guojun Yin, Luchuan Song, Lu Sheng, Jing Shao, and Ziwei Liu. Forgerynet: A versatile benchmark for comprehensive forgery analysis. *arXiv preprint arXiv:2103.05630*, 2021. 1, 3, 2
- [19] Todd C. Helmus. Artificial intelligence, deepfakes, and disinformation: A primer. Technical report, RAND Corporation, 2022. 2
- [20] Todd C. Helmus. Artificial intelligence, deepfakes, and disinformation: A primer. Technical report, RAND Corporation, 2022. 1
- [21] iProov. Threat intelligence report 2024: The impact of generative AI on remote identity verification. Technical report, iProov, 2024. 4
- [22] Liming Jiang, Ren Li, Wayne Wu, Chen Qian, and Chen Change Loy. Deepforensics-1.0: A large-scale dataset for real-world face forgery detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2889–2898, 2020. 7
- [23] Liming Jiang, Ren Li, Wayne Wu, Chen Qian, and Chen Change Loy. DeepForensics-1.0: A large-scale dataset for real-world face forgery detection. In *CVPR*, 2020. 1, 3, 2
- [24] Jee-weon Jung, Hee-Soo Heo, Hemlata Tak, Hye-jin Shim, Joon Son Chung, Bong-Jin Lee, Ha-Jin Yu, and Nicholas Evans. Aassist: Audio anti-spoofing using integrated spectro-temporal graph attention networks. In *arXiv preprint arXiv:2110.01200*, 2021. 1, 5, 7
- [25] Jee-weon Jung, Yihan Wu, Xin Wang, Ji-Hoon Kim, Soumi Maiti, Yuta Matsunaga, Hye-jin Shim, Jinchuan Tian, Nicholas Evans, Joon Son Chung, et al. Spoofceleb: Speech deepfake detection and sasv in the wild. *IEEE Open Journal of Signal Processing*, 2024. 1, 3
- [26] N. Kamali, K. Nakamura, A. Chatzimpampas, J. Hullman, and M. Groh. How to distinguish AI-generated images from authentic photographs. *arXiv preprint arXiv:2406.08651*, 2024. 4, 7, 8
- [27] Hasam Khalid, Shahroz Tariq, and Simon S. Woo. Fakeavceleb: A novel audio-video multimodal deepfake dataset, 2021. 1, 2, 3
- [28] Davis E. King. Dlib-ml: A machine learning toolkit. *J. Mach. Learn. Res.*, 10:1755–1758, 2009. 4
- [29] Pavel Korshunov and Sebastien Marcel. Deepfakes: a new threat to face recognition? assessment and detection, 2018. 7
- [30] Lingzhi Li, Jianmin Bao, Hao Yang, Dong Chen, and Fang Wen. Faceshifter: Towards high fidelity and occlusion aware face swapping. *arXiv preprint arXiv:1912.13457*, 2019. 7

- [31] Yuezun Li, Xin Yang, Pu Sun, Honggang Qi, and Siwei Lyu. Celeb-df: A large-scale challenging dataset for deepfake forensics. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020. 7, 2
- [32] Yewon Lim, Changyeon Lee, Aerin Kim, and Oren Etzioni. Distildire: A small, fast, cheap and lightweight diffusion synthesized deepfake detection. *arXiv preprint arXiv:2406.00856*, 2024. 5, 7
- [33] Xuechen Liu, Xin Wang, Md Sahidullah, Jose Patino, Héctor Delgado, Tomi Kinnunen, Massimiliano Todisco, Junichi Yamagishi, Nicholas Evans, Andreas Nautsch, and Kong Aik Lee. Asvspoof 2021: Towards spoofed and deepfake speech detection in the wild. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 31:2507–2522, 2023. 1, 3
- [34] Govind Mittal, Chinmay Hegde, and Nasir Memon. Gotcha: Real-time video deepfake detection via challenge-response, 2023. 2
- [35] Nicolas M Müller, Pavel Czempin, Franziska Dieckmann, Adam Froghyar, and Konstantin Böttinger. Does audio deepfake detection generalize? *Interspeech*, 2022. 1, 2, 3
- [36] Kartik Narayan, Harsh Agarwal, Kartik Thakral, Surbhi Mittal, Mayank Vatsa, and Richa Singh. Df-platter: Multi-face heterogeneous deepfake dataset. In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9739–9748, 2023. 2
- [37] João C. Neves, Ruben Tolosana, Ruben Vera-Rodriguez, Vasco Lopes, Hugo Proença, and Julian Fierrez. Ganprintr: Improved fakes and evaluation of the state of the art in face manipulation detection. *IEEE Journal of Selected Topics in Signal Processing*, 14(5):1038–1048, 2020. 3
- [38] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. Glide: Towards photorealistic image generation and editing with text-guided diffusion models, 2022. 7
- [39] Utkarsh Ojha, Yuheng Li, and Yong Jae Lee. Towards universal fake image detectors that generalize across generative models. In *CVPR*, 2023. 5, 6, 7
- [40] Jordan Peele. Deep fake of barack obama, 2018. A public service announcement video demonstrating deepfake technology by having Jordan Peele’s voice and lip movements digitally mapped onto footage of former President Barack Obama. 5
- [41] Jiameng Pu, Neal Mangaokar, Lauren Kelly, Parantapa Bhat-tacharya, Kavya Sundaram, Mobin Javed, Bolun Wang, and Bimal Viswanath. Deepfake videos in the wild: Analysis and detection. In *Proceedings of The Web Conference 2021*, 2021. 2
- [42] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021. 5
- [43] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision, 2022. 4
- [44] Ricardo Reimao and Vassilios Tzerpos. For: A dataset for synthetic speech detection. *2019 International Conference on Speech Technology and Human-Computer Dialogue (SpED)*, pages 1–10, 2019. 3
- [45] Aja Romano. Jordan peele’s simulated obama psa is a double-edged warning against fake news. *Vox*, 2018. Accessed May 14, 2025. 5
- [46] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022. 1, 7
- [47] Andreas Rössler, Davide Cozzolino, Luisa Verdoliva, Christian Riess, Justus Thies, and Matthias Nießner. FaceForensics++: Learning to detect manipulated facial images. In *International Conference on Computer Vision (ICCV)*, 2019. 1, 3, 7, 2
- [48] Christoph Schuhmann, Richard Vencu, Romain Beaumont, Robert Kaczmarczyk, Clayton Mullis, Aarush Katta, Theo Coombes, Jenia Jitsev, and Aran Komatsuzaki. Laion-400m: Open dataset of clip-filtered 400 million image-text pairs, 2021. 7
- [49] Snopes Media Group Inc. About snopes, 2025. Snopes is a widely trusted fact-checking website that employs professional journalists who investigate viral claims and urban legends. 4
- [50] Sum and Substance Ltd. Identity fraud report 2024. Technical report, Sum and Substance Ltd., 2024. 1
- [51] Supasorn Suwajanakorn, Steven M Seitz, and Ira Kemelmacher-Shlizerman. Synthesizing obama: learning lip sync from audio. *ACM Transactions on Graphics (ToG)*, 36(4):1–13, 2017. 1
- [52] Hemlata Tak, Jose Patino, Massimiliano Todisco, Andreas Nautsch, Nicholas Evans, and Anthony Larcher. End-to-end anti-spoofing with rawnet2, 2021. 5, 7
- [53] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, and Yunchao Wei. Rethinking the up-sampling operations in cnn-based generative network for generalizable deepfake detection, 2023. 1, 5, 7
- [54] Chuangchuang Tan, Renshuai Tao, Huan Liu, and Yao Zhao. Gangen-detection: A dataset generated by gans for generalizable deepfake detection. <https://github.com/chuangchuangtan/GANGen-Detection>, 2024. 7
- [55] Khang-Christopher Duc Truong. Reputation (not taylor’s version): Regulating artificial intelligence hallucinated deepfakes of public figures. *Illinois Journal of Law, Technology ‘I&’ Policy*, 2024(2), 2024. 1
- [56] Rebecca Umbach, Nicola Henry, Gemma Beard, and Colleen Berryessa. Non-consensual synthetic intimate imagery: Prevalence, attitudes, and knowledge in 10 countries, 2024. 1
- [57] Sheng-Yu Wang, Oliver Wang, Richard Zhang, Andrew Owens, and Alexei A Efros. Cnn-generated images are surprisingly easy to spot...for now. In *CVPR*, 2020. 7, 3
- [58] Xin Wang and Junichi Yamagishi. Can large-scale vocoded spoofed data improve speech spoofing countermeasure with a self-supervised front end?, 2023. 5, 7

- [59] Xin Wang, Junichi Yamagishi, Massimiliano Todisco, Hector Delgado, Andreas Nautsch, Nicholas Evans, Md Sahidullah, Ville Vestman, Tomi Kinnunen, Kong Aik Lee, Lauri Juvela, Paavo Alku, Yu-Huai Peng, Hsin-Te Hwang, Yu Tsao, Hsin-Min Wang, Sebastien Le Maguer, Markus Becker, Fergus Henderson, Rob Clark, Yu Zhang, Quan Wang, Ye Jia, Kai Onuma, Koji Mushika, Takashi Kaneda, Yuan Jiang, Li-Juan Liu, Yi-Chiao Wu, Wen-Chin Huang, Tomoki Toda, Kou Tanaka, Hirokazu Kameoka, Ingmar Steiner, Driss Matriouf, Jean-Francois Bonastre, Avashna Govender, Srikanth Ronanki, Jing-Xuan Zhang, and Zhen-Hua Ling. Asvspoof 2019: A large-scale public database of synthesized, converted and replayed speech, 2020. [7](#), [3](#)
- [60] Xin Wang, Héctor Delgado, Hemlata Tak, Jee-weon Jung, Hye-jin Shim, Massimiliano Todisco, Ivan Kukanov, Xuechen Liu, Md Sahidullah, Tomi Kinnunen, et al. Asvspoof 5: Crowdsourced speech data, deepfakes, and adversarial attacks at scale. *arXiv preprint arXiv:2408.08739*, 2024. [6](#)
- [61] Zhendong Wang, Jianmin Bao, Wengang Zhou, Weilun Wang, Hezhen Hu, Hong Chen, and Houqiang Li. Dire for diffusion-generated image detection. *arXiv preprint arXiv:2303.09295*, 2023. [3](#), [7](#)
- [62] Deressa Wodajo, Solomon Atnafu, and Zahid Akhtar. Deepfake video detection using generative convolutional vision transformer, 2023. [1](#), [5](#), [7](#)
- [63] Junichi Yamagishi, Massimiliano Todisco, Md Sahidullah, Héctor Delgado, Xin Wang, Nicholas Evans, Tomi Kinnunen, K Aik Lee, Ville Vestman, and Andreas Nautsch. Asvspoof 2019: Automatic speaker verification spoofing and countermeasures challenge evaluation plan. *ASV Spoof*, 13, 2019. [3](#)
- [64] Zhiyuan Yan, Taiping Yao, Shen Chen, Yandan Zhao, Xinghe Fu, Junwei Zhu, Donghao Luo, Li Yuan, Chengjie Wang, Shouhong Ding, et al. Df40: Toward next-generation deepfake detection. *arXiv preprint arXiv:2406.13495*, 2024. [6](#), [2](#)
- [65] Jiangyan Yi, Chenglong Wang, Jianhua Tao, Xiaohui Zhang, Chu Yuan Zhang, and Yan Zhao. Audio deepfake detection: A survey, 2023. [3](#)
- [66] Yinglin Zheng, Jianmin Bao, Dong Chen, Ming Zeng, and Fang Wen. Exploring temporal coherence for more general video face forgery detection, 2021. [5](#), [7](#)
- [67] Bojia Zi, Minghao Chang, Jingjing Chen, Xingjun Ma, and Yu-Gang Jiang. Wilddeepfake: A challenging real-world dataset for deepfake detection. In *Proceedings of the 28th ACM International Conference on Multimedia*, pages 2382–2390, 2020. [3](#), [2](#)